Parity Panic: A Binary Game

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***Abstract*— In recent years, interactive tools have proven highly effective in engaging students, especially in STEM fields, by making learning more dynamic and practical. Thus, the implementation of "Parity Panic", a game where binary numbers are identified as odd or even, is designed. The project aims to enhance the learning experience in the field of Computer Science and Electronics by utilizing the base two numeral and embedded system. Its creation employs an Arduino-based system with hardware components. Thus, including an ultrasonic sensor, buttons, LCD I2C display, LED indicators and auditory cues for real-time feedback. Guided by its algorithm, Parity Panic integrates each hardware to produce a seamless game flow. To enhance player engagement, the game features difficulty modes that adjust the timer dynamically across three levels - easy, medium, hard. As the user progress, the challenge is increased as it shifts from a 4-bit binary length to an 8-bit. Results have shown that by applying the various software and hardware components, player interaction and engagement is enhanced. However, improvement in feedback mechanism and high-score tracking is highly suggested to further increase engagement and provide learners with a competitive and captivating learning tool.

***Index Terms*— binary game, Arduino, embedded systems, prototype, educational games, interactive educational tool**

I. Introduction

Interactive educational tools have been developed and used in several course and subject areas specially in science, technology, engineering, and mathematics (STEM) courses allowing great student engagement and learning outcomes. Various study has utilized educational games in education for more interactive and engaging learning [3][4]. A study shows that hands-on, entertaining approach enhance student comprehension and retention around certain concept and topics, especially in fields like computer science and electronics [1][2][4]. With this, binary number systems are one of the fundamental concepts for computer and electronics courses underlining the logic of all digital computation. With advancement of educational tools, using a game to fully engage and give practical application and real-time feedback would allow students it's fun and entertaining learning experience.

Therefore, Parity Panic introduced educational and interactive game design to engage player to a quick, critical thinking about binary and decimal number. Parity Panic offers a gaming experience where player can practice their binary-to-decimal conversion, making odd/even decision under timer pressure. The game run an Arduino-based system with ultrasonic sensor for player detection, an LCD I2C 20x4 for display, and LED lights with arcade music to create an engaging experience.

Designed for all learner, the game has 3 modes of difficulty: easy, medium, and hard adjusting the timer in each mode. together with its applications in embedded systems, the gameplay mechanism and real-time feedback will make learning possible and sharpen decision-making skills and critical thinking. Besides this, Parity Panic, in demonstrating practical concepts of embedded systems through an Arduino kit and simple handshaking protocols, provides it as an excellent introductory tool in both binary arithmetic and microcontroller-based projects.

In this paper, we review the design and implementation of Parity Panic: The Binary Game when it comes to hardware, scoring mechanism, user interface, and game progression. We explore its potential to be an educational tool as a testing of concept tool about the foundational computing concepts.

II. Methodology

The Parity Panic: A Binary Game will be developed by applying concepts in embedded system and design.

A. Hardware Development

Parity Panic hardware components consist of easily accessible component allowing easy recreation and development. Parity panic utilizes an ultrasonic for player detection and two input buttons for answer, and another 3 inputs for changing of mode and starting the game, as shown in the hardware block diagram in Fig 1. The output components include LED indicator for lives, LCD I2C 20x4 to display the game, and a buzzer to provide auditory feedback.

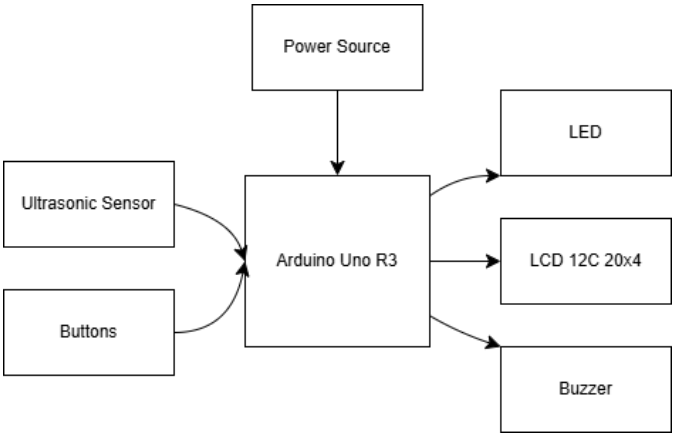

Fig. 1 Hardware block diagram of Parity Panic

When a player presence is detected within the ultrasonic distance threshold, the game display will start, achieving objective 2 wherein the system will sense the presence of a player before the display is activated. LCD screen presents the overall user interface, interfacing player score, time, binary, and player status. The two buttons for odd and even are the interaction of the players with the game which are processed in Arduino uno for gameplay decisions.

LED lights are also an output component which are use as a life indicator of the system, when an incorrect response is made, one turns off, leaving the player visual notice of how many attempts remain. The buzzer sounds during various states of play. This would make the game that much more arcade experience-oriented for the player and immediately provides auditory feedback.

B. Software Development

Fig. 2 illustrates the main flowchart for the software development of the system. the process begins with initialization of all components and sensors. It employs the use of the ultrasonic sensor in detecting a player by checking whether there is someone standing there and ready to play. The startup screen is then displayed along with a game mode to be played afterward. In the case where the button for mode is pressed, it will register the selected mode. Pressing the start button initializes the game, which starts a renewal cycle for the timer, score, level, and the binary generator. In case either of the lives or the timer reaches zero, then a game-over screen is displayed. Otherwise, if player is able to reach level 3 and complete 10 sets, it displays a congratulatory message.

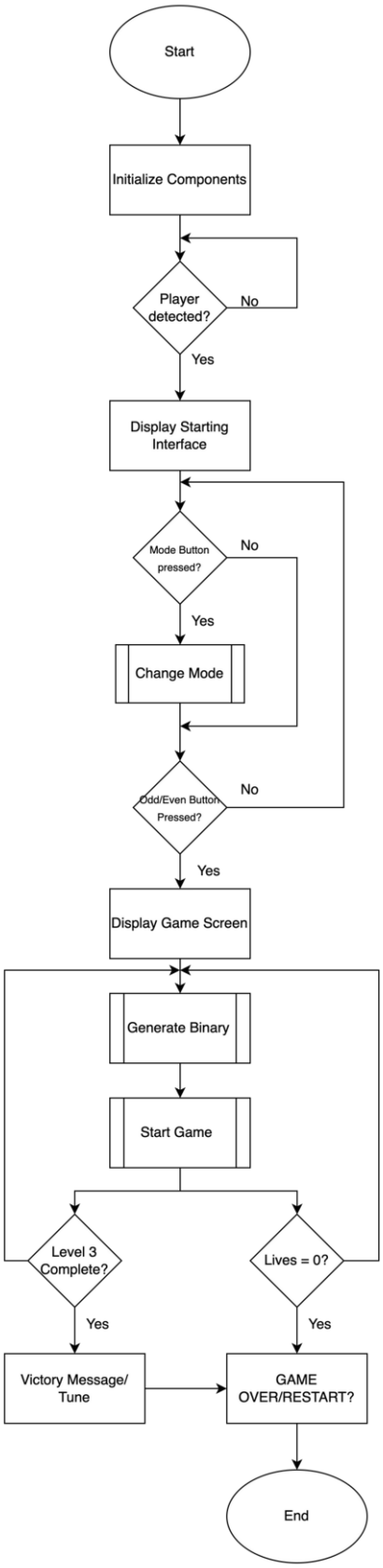

Fig. 2 Main flowchart of the software development of the Parity Panic

Player has the freedom to change mode in the display startup interface. Fig 3. Illustrates the flow of overall change of mode function. The flow begins with change of mode action, implying that the player opted to switch difficulty levels. The system checks if the game is currently in Easy or Medium mode, Easy mode switches to Medium, Medium to Hard, and Hard loops back to Easy. This loop updates the game's mode setting and applies the selected difficulty when the game starts. Return action indicate that the player returned to the main menu keeping mode selection as a distinct, standalone process that doesn't interfere with other game functions.

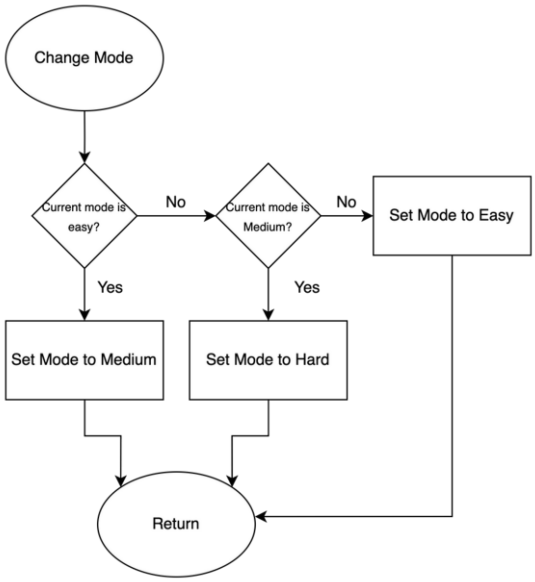

Fig. 3 Flowchart of change mode

Fig. 4 illustrates the gameplay loop and core mechanism. The game starts by displaying the initial binary number together with time, level, and score. Countdown timer then will start pressuring player to have quick decision of answer. Once player push "odd" or "even" button the program will check if the answer is right or wrong. When the player answer is wrong, the initialize lives of 3 decrement then the game loops back into another set of binary number. If the answer is correct, the score is incremented, and the display is updated with the new score and level. A new binary number is generated, and the game continues with the updated timer. The cycles are repeated until the player's lives go to zero; then the game would return to the main menu, or a "game over" message would appear. This loop structure creates continuous play, concurrently tracking and updating the score, number of lives, and the timer, making things challenging and interesting.

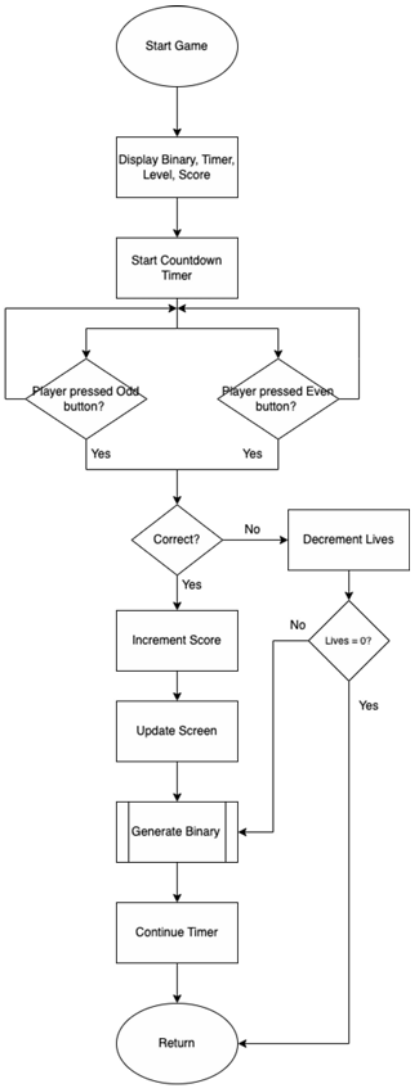

Fig. 4 Flowchart of start game

Fig 5. Random Generate Binary illustrated the logic for generating the length of binary number per level. The process begins with command to generate a binary number, then checks the current level. When the current level reaches 1, the system creates a 4-bit binary number, thus delivering a less challenging for beginner players. If the current level is 2, it generates 6-bit binary number and becomes hard for the player to memorize a longer sequence. In the case of a level 3 player, an 8-bit binary number is also created to denote the highest amount of difficulty with the longest sequence of the binary number. This approach ensures that the game progresses with escalating difficulty, and thus, the challenge for the player will also increase as they progress in levels. Once the right binary number is created, the game loop is reintroduced, displaying the binary number across the screen, and now, the game continues into its new parameters. This process maintains a balanced escalatory increasing level of difficulty as it offers increasingly difficult tasks for the player to complete.

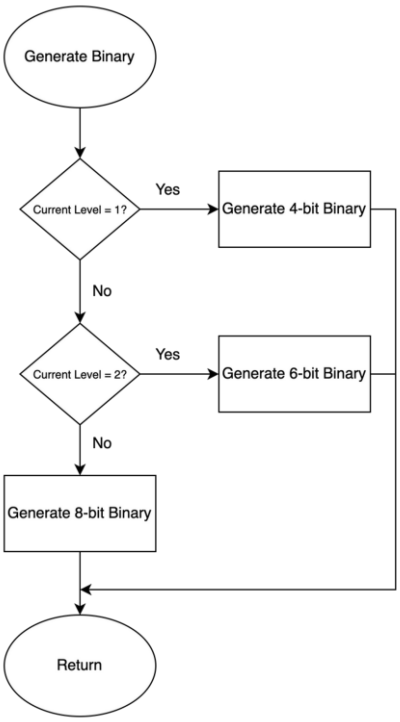

Fig. 5 Flowchart for random generate binary

C. *Casing*

All components mentioned including the Arduino uno Is safely place and mounted on the created casing using foam board. The buttons and led is placed in 2 breadboards connected to each other. The lcd is glued to the board to stay in place throughout its journey while ultrasonic sensor is mounted at the top of the casing. This casing is inspired in with a Gameboy casing concept. Creating nostalgic and adventurous looking casing.

Fig. 6 Parity Panic Casing

III. Results and Discussion

The prototype was successfully implemented using the flowchart presented in Chapter II. Using the guided flowchart, it creates seamless flow of function enabling smooth updates and connection between different variables and component. Fig 7. Shows snippet of code in the check answer function.

The design test users in their knowledge about accurately identifying binary number as either odd or even, using techniques such as tactile feedback, auditory cues, and visual indicators. With the Arduino based system paired with LCD, ultrasonic sensor, LED, and buzzer, the game presents an immersive experience with multiple modes of difficulty, dynamic scoring, and real time feedback.

Ultrasonic sensor enhanced the interactivity of the game determining the distance of the player within the device. With this information the game could only be activated when the player is at front 100cm ahead. Fig 8. Shows the ultrasonic setup where game interface is at pause as player is out of reach. LED indicators and buzzer were used as a lives indicator, wherein initial lives of a player is 3 and wrong answer would turn off the first led light. Fig 9. Shows the game over once lives are equal to 0. Buzzer is used as background music for more entertaining game experience.

The system had dynamic levels of difficulties as the length of the binary number increased and so does the time per round. The LCD screen was the most essential in showing the current level, remaining time, and the binary number to be categorized, updating every 1s. Fig 10. Shows the in-game display and Fig 11. Shows the congratulation display.

```
void checkAnswer(bool playerAnswer) {
    // Compare player answer with actual binary
    if ((currentBinary % 2 == 0 && playerAnswer) || (currentBinary % 2 != 0 && !playerAnswer)) {
        streak++;

        // Base score depending on streak
        int baseScore = 0;
        if (streak == 1) {
            baseScore = 5;  // Base score for the first streak
        } else if (streak == 2) {
            baseScore = 10; // Base score for the second streak
            lcd.setCursor(15, 3);
            lcd.print("NICE");
        } else {
            baseScore = 15; // Base score for the third or more streaks
            lcd.setCursor(14, 3);
            lcd.print("Combo!");
        }

        // Modify score based on selected mode
        if (mode == 0) {
            // Easy mode, base score stays the same
            score += baseScore;
        } else if (mode == 1) {
            // Medium mode, 20% higher score
            score += baseScore * 1.2;
        } else if (mode == 2) {
            // Hard mode, 40% higher score
            score += baseScore * 1.4;
        }

    } else {
        lives--;
        streak = 0;
        lcd.setCursor(14, 3);
        lcd.print("      ");

        // Update LEDs based on lives count
```

Fig. 7 Code snippet for check answer

Fig. 8 Ultrasonic Setup with game interface at pause

Fig. 9 Game Over Display

Fig. 10 In Game Display

Fig. 11 Congratulation Display

Problems encountered while implementing the whole system includes timing issues, ultrasonic sensor calibration, and proper creation of algorithm logic. Handling the timer countdown per set of binaries is the hardest to implement without proper delay and coordination between different variables. At one point in the process, the timing code causes problem in checking answers and updating lcd gameplay therefore optimizing and retracking of step is essential to fix the problem. Nonetheless, once timing operation is achieved algorithm of checking and scoring is created more seamlessly than creation of timing. Streak system and modes are coordinated to each other for more complex scoring within the game. Fig. 12 shows the times up interface once player didn't answer within the time limit. For the handling of ultrasonic sensor, as sometimes the measurement of distance fluctuates, handling its fluctuation by running the function of pause only if the distance is measured less than 100cm within 2.5 seconds.

Fig. 12 Time's Up Display

Overall, code optimization with hardware integration were needed to make the "Parity Panic" game successful. Although the system was working as expected, additional enhancements from the sound effects to the difficulty progression of the game will make it even better to experience.

IV. Conclusions and Recommendations

In Conclusion, Parity Panic demonstrate the capabilities of developing interactive educational games to enhance computer science and electronics education. This project can capture good integration that encapsulates the concepts of binary numbers in integration with embedded systems, creating engaging platform that supports critical thinking real time. The system employs an Arduino-based setup with an LCD I2C display, ultrasonic sensor, LED indicators, buttons, and auditory cues which altogether enhance the user experience through learning by interactive gaming. The overall approach of this project toward the game difficulty-easily built up-from four bit binary to eight bits binary-again through each challenge-it is meant to present a progressive challenge, progressively building the player's binary-to-decimal conversion skills through practice. This game effectively achieves smooth gameplay and reliable scoring mechanisms through troubleshooting in timing control, ultrasonic sensor calibration, and code optimization. Such aspects underscore the project's contribution in giving students meaningful tools in computer concept education.

To further enhance its whole system, improvement of feedback mechanism such as auditory cues could furthermore enhance the player gameplay. Additionally, creating more complex scoring and previous high score will allow the system to create more competitive nature to attract player in continues gameplay. Moreover, using a more compact, robust casing design would enhance portability and durability, making Parity Panic more accessible for educational settings.

References

[1] A. Rojas-Salazar and M. Haahr, "The Effect of Educational Video Game Perceptual Realism on Learning Outcomes and Motivation," 2024 IEEE Gaming, Entertainment, and Media Conference (GEM), Turin, Italy, 2024, pp. 1-6, doi: 10.1109/GEM61861.2024.10585393.

[2] Youngkwon Bae, Jinsook Lim and Taewuk Lee, "Work in progress - a study on educational computer games for e-learning based on activity theory," Proceedings Frontiers in Education 35th Annual Conference, Indianopolis, IN, USA, 2005, pp. F1C-18, doi: 10.1109/FIE.2005.1612004.

[3] Y. -S. Chou et al., "Running Tommy©: Developing a Digital Adventure Game Based on Situated Learning to Promote Learners' Concepts of Earthquake Escape," 2012 IEEE Fourth International Conference On Digital Game And Intelligent Toy Enhanced Learning, Takamatsu, Japan, 2012, pp. 156-158, doi: 10.1109/DIGITEL.2012.44.

[4] Z. Karapinar, A. Senturk, S. Zavrak, R. Kara and P. Erdogmus, "Binary apple tree: A game approach to tree traversal algorithms," 2012 International Conference on Information Technology Based Higher Education and Training (ITHET), Istanbul, Turkey, 2012, pp. 1-3, doi: 10.1109/ITHET.2012.6246026.